Cross-Resolution Consistency Learning for Robust Bean Disease Classification Using CNN, ResNet, and Vision Transformer Architectures

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1. INTRODUCTION

Deep learning has significantly advanced the field of image classification, achieving remarkable performance across a wide range of real-world applications. However, a key limitation of many deep learning models is their sensitivity to variations in image resolution and scale. Models trained on high-resolution images often fail to generalize effectively when presented with lower-resolution inputs, resulting in inconsistent and unreliable predictions. This issue becomes critical in practical scenarios where image quality may vary due to differences in camera devices, environmental conditions, or preprocessing techniques [1].

To address this challenge, this project explores Cross-Resolution Consistency Learning, a technique designed to improve the robustness of image classification models across multiple scales. Instead of training on a single resolution, each input image is transformed into multiple versions at different resolutions. The model is then trained using two objectives, a standard classification loss to ensure prediction accuracy, and a consistency loss based on Kullback-Leibler divergence, which enforces similar predictions across different resolutions. This encourages the model to learn scale-invariant features and improves its ability to generalize across varying input conditions [2].

The performance of this approach is evaluated using three widely used deep learning architectures: a VGG-style Convolutional Neural Network (CNN), a Residual Network (ResNet), and a Vision Transformer (ViT). VGG networks demonstrate the effectiveness of deep convolutional architectures [3], while ResNet introduces residual learning to address the degradation problem in deep networks [4]. More recently, Vision Transformers have shown strong performance by leveraging self-attention mechanisms to capture global dependencies in images [5].

The experiments are conducted on a bean disease classification dataset obtained from AI-Lab Makerere in collaboration with the National Crops Resources Research Institute (NaCRRI), Uganda. The dataset consists of real-world images captured in agricultural fields and includes three classes: Angular Leaf Spot, Bean Rust, and Healthy leaves [6]. By evaluating both classification accuracy and prediction consistency across resolutions, this project highlights the importance of robustness in deep learning models and demonstrates how cross-resolution learning can enhance reliability in practical applications.

2. LITERATURE REVIEW

Recent advancements in deep learning have increasingly focused on improving the robustness of image classification models under varying resolutions and scales. Traditional models often exhibit performance degradation when tested on images with resolutions different from those seen during training, highlighting the need for scale-invariant learning mechanisms.

Zhang *et al.* (2026) proposed a meta-weight learning framework for continuous-scale image super-resolution, demonstrating that adaptive feature aggregation can significantly enhance cross-resolution generalization and reconstruction quality [7]. Extending this idea, Mantes *et al.* (2026) introduced a multi-resolution vision transformer that jointly learns representations across different scales using masked reconstruction, enabling improved feature alignment and robustness [8]. Park *et al.* (2025) proposed a visually consistent hierarchical classification framework that enforces prediction consistency across different semantic levels, leading to improved classification accuracy and interpretability [9]. Similarly, Tao *et al.* (2025) introduced a multi-task learning approach with cross-task consistency constraints, showing that consistency regularization can enhance semantic coherence and overall model performance [10].

Consistency learning has also been explored in multi-scale and semi-supervised settings. Liu *et al.* (2026) proposed a dual-consistency learning framework for image segmentation, where consistency is enforced between coarse and fine representations, resulting in more stable predictions [11]. Likewise, Zhai *et al.* (2023) developed a weakly supervised self-consistency learning method that leverages inter-patch relationships to improve generalization under distribution shifts [12]. In the domain of low-resolution recognition, Saadabadi *et al.* (2023) introduced a classification consistency distillation framework for low-resolution face recognition. Their approach transfers knowledge from high-resolution to low-resolution representations, improving discriminative performance and robustness [13].

Multi-scale feature learning has also been widely studied in transformer-based architectures. Chen *et al.* (2021) proposed CrossViT, a cross-attention-based multi-scale vision transformer that effectively integrates features from different patch sizes, enhancing classification performance [14]. More recently, Sankari *et al.* (2025) demonstrated that hierarchical multi-scale vision transformers can capture both local and global dependencies, leading to improved classification outcomes in complex datasets [15]. Additionally, Lai *et al.* (2025) presented a comprehensive study on integrating attention mechanisms with multi-scale learning, concluding that such integration significantly improves feature representation and model robustness across varying image resolutions [16].

Overall, the literature strongly indicates that incorporating multi-scale representations and enforcing prediction consistency across resolutions are effective strategies for improving robustness and reliability in image classification models.

3. METHODOLOGY

This study employs deep learning-based architectures to address the problem of resolution sensitivity in image classification. Specifically, three representative models are implemented and comparatively evaluated: a VGG-style Convolutional Neural Network (CNN), a Residual Network (ResNet), and a Vision Transformer (ViT). In addition to these architectures, a cross-resolution consistency learning framework is incorporated to enforce prediction stability across multiple image scales, aligning with recent advancements in robust visual representation learning.

3.1. VGG-Style Convolutional Neural Network

The VGG-style CNN is designed to capture hierarchical spatial features through stacked convolutional operations. The architecture consists of multiple convolutional blocks, each comprising convolutional layers with small receptive fields (3×3), followed by batch normalization and non-linear activation. Max-pooling layers are employed to progressively reduce spatial dimensions, while an adaptive average pooling layer ensures compatibility with varying input resolutions. The extracted features are passed through fully connected layers with dropout regularization to produce the final classification output.

3.2. Residual Network (ResNet)

To address the degradation problem associated with deep networks, a simplified ResNet architecture is utilized. The model incorporates residual blocks with identity skip connections, allowing gradients to propagate effectively during backpropagation. Each residual block consists of convolutional layers, batch normalization, and ReLU activation, with shortcut connections enabling feature reuse. Downsampling is performed using strided convolutions, followed by global average pooling and a fully connected classification layer.

3.3. Vision Transformer (ViT)

The Vision Transformer represents a paradigm shift from convolutional processing to attention-based modeling. The input image is divided into fixed-size patches, which are linearly embedded into a sequence of tokens. Positional embeddings are added to retain spatial information. The token sequence is processed through multiple transformer encoder layers, each consisting of multi-head self-attention and feed-forward networks. A learnable classification token aggregates global information, and its representation is used for final prediction. An adaptive resizing mechanism is incorporated to standardize inputs across varying resolutions.

3.4. Activation Functions

Rectified Linear Unit (ReLU) activation is employed in convolutional and residual layers to introduce non-linearity and enable efficient gradient propagation. At the output layer, the Softmax function is used to convert logits into class probabilities.

3.5. Loss Function

The training objective combines classification and consistency constraints. The primary loss function is categorical cross-entropy, which measures the discrepancy between predicted and true labels. To enforce scale-invariant predictions, a consistency loss based on Kullback-Leibler (KL) divergence is introduced. This term penalizes discrepancies between the predicted probability distributions obtained from different image resolutions.

The total loss is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{consistency}$$

where λ controls the relative importance of the consistency term.

3.6. Optimizer

Model parameters are optimized using the Adam optimizer, which provides adaptive learning rates and efficient convergence. A step-based learning rate scheduler is employed to reduce the learning rate periodically, thereby improving training stability and preventing overfitting.

3.7. Workflow

The proposed framework follows a systematic pipeline comprising data preparation, model training, and evaluation:

1. ***Data Acquisition***

   The dataset consists of bean leaf images categorized into three classes: Angular Leaf Spot, Bean Rust, and Healthy.

2. ***Preprocessing***

   Images are resized into multiple resolutions (224×224, 128×128, and 64×64, and 32×32) to simulate scale variation. Data augmentation techniques, including random cropping, horizontal flipping, and color jittering, are applied to improve generalization. All images are normalized using standard ImageNet statistics.

3. ***Training Phase***

   During training, each input image is processed at multiple resolutions. The model generates predictions for each scale, and the combined loss function (classification + consistency) is minimized backpropagation.

4. ***Testing Phase***

The trained models are evaluated on unseen test data. Predictions are primarily obtained from the highest resolution input while consistency across resolutions is also analyzed.

5. ***Evaluation***

Model performance is assessed using classification accuracy, confusion matrices, and a consistency score that quantifies prediction agreement across resolutions.

4. HYPERPARAMETER SETTING

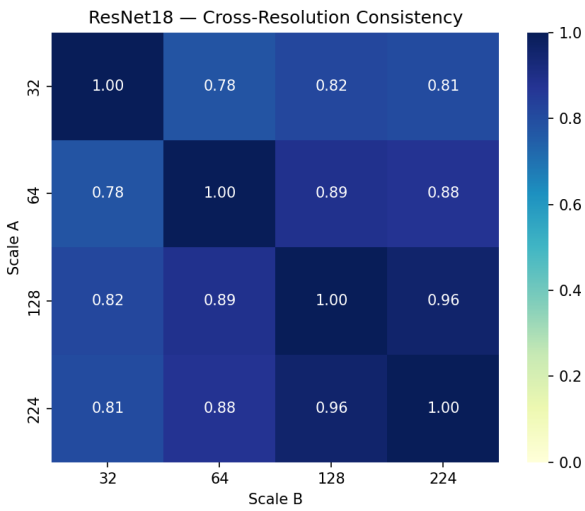
The following hyperparameters are used throughout the experiments:

Hyperparameter	Value
Batch Size	32
Number of Epochs	15
Learning Rate	0.001
Optimizer	Adam
Consistency Weight (λ)	0.3
Input Resolutions	224, 128, 64, 32
Number of Classes	3
Dropout Rate	0.5

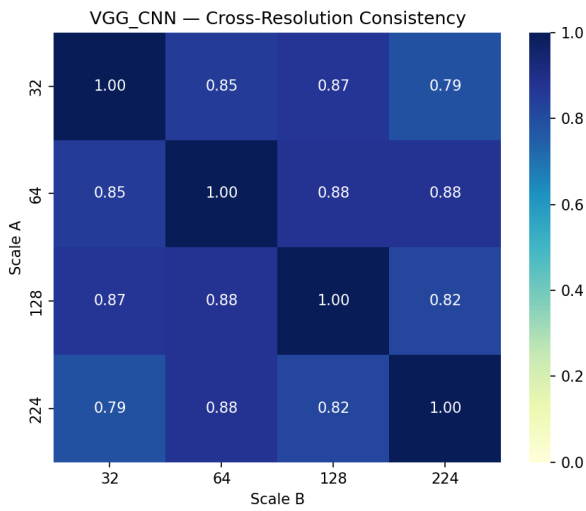
5. RESULT ANALYSIS

5.1. Cross Resolution Consistency

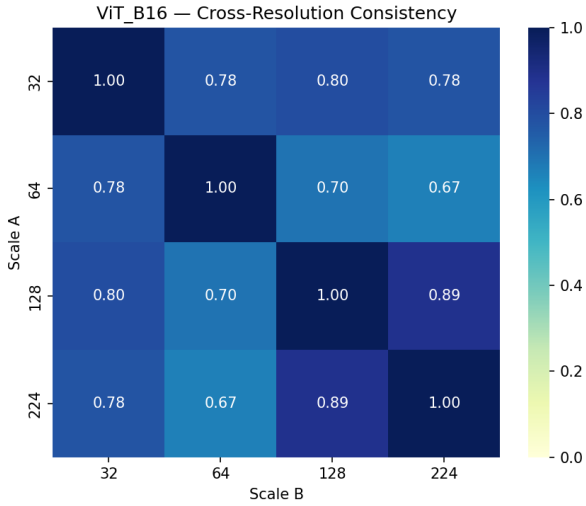
Figure 1 illustrates the cross-resolution consistency achieved by three different models, ResNet18, VGG-style CNN, and ViT-B16, across varying image scales. As observed in Fig. 1, both ResNet18 and VGG exhibit relatively high consistency scores across most resolution pairs, indicating stable predictions when the input scale changes. In particular, VGG demonstrates slightly more uniform consistency across all scales, while ResNet18 shows strong agreement at higher resolutions (128-224). In contrast, ViT-B16 displays comparatively lower consistency, especially between mid and lower resolutions (e.g., 64-128 and 64-224), suggesting sensitivity to scale variations. The diagonal values in all heatmaps remain at 1.0, representing perfect self-consistency at the same resolution. Overall, Fig. 1 highlights that CNN-based architectures tend to maintain better cross-resolution stability compared to transformer-based models in this setup.



(a) ResNet18



(a) VGG

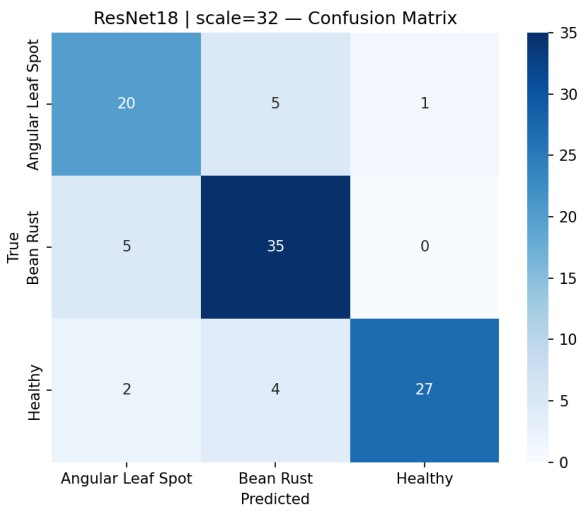


(c) ViT_B16

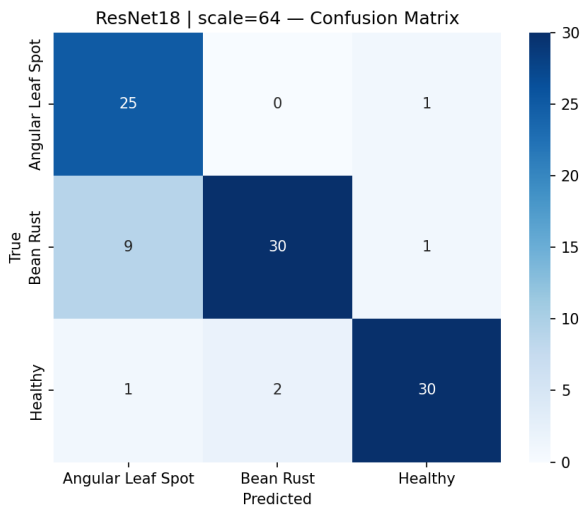
Fig. 1 Cross Resolution consistency for three models *viz.* (a) ResNet18, (b) VGG, and (c) ViT_B16

5.2. Confusion Matrix

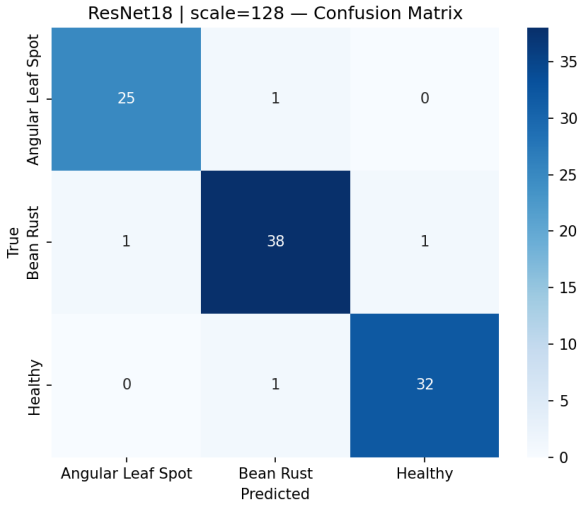
Figure 2 presents the confusion matrices of the ResNet18 model evaluated at different input resolutions (32, 64, 128, and 224). As observed in Fig. 2, the classification performance improves progressively with increasing resolution, with higher diagonal values indicating more correct predictions. At lower resolutions (32 and 64), the model exhibits noticeable misclassifications, particularly between Angular Leaf Spot and Bean Rust. However, at higher resolutions (128 and 224), the confusion significantly reduces, and the predictions become more accurate and stable across all classes. The model achieves near-perfect classification for the Healthy class at higher scales, while Bean Rust also shows strong performance. Overall, Fig. 2 demonstrates that increasing image resolution enhances feature representation, leading to improved classification accuracy and reduced inter-class confusion.



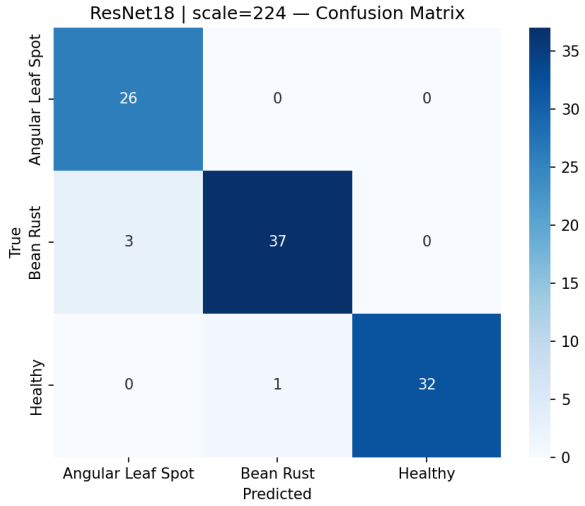
(a) ResNet18_32



(a) ResNet18_64



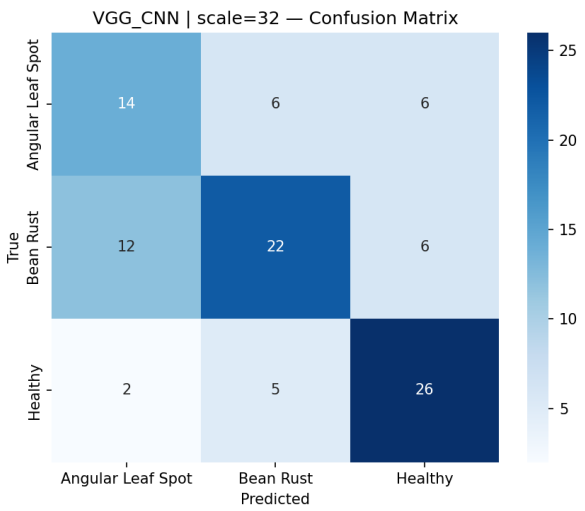
(c) ResNet18_128



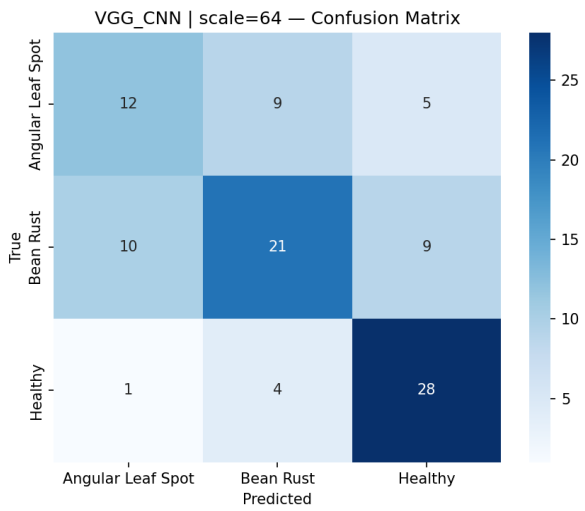
(d) ResNet18_224

Fig. 2 Confusion Matrix for ResNet at different scales (a) ResNet18_32, (b) ResNet18_64, (c) ResNet18_128, and (d) ResNet18_224

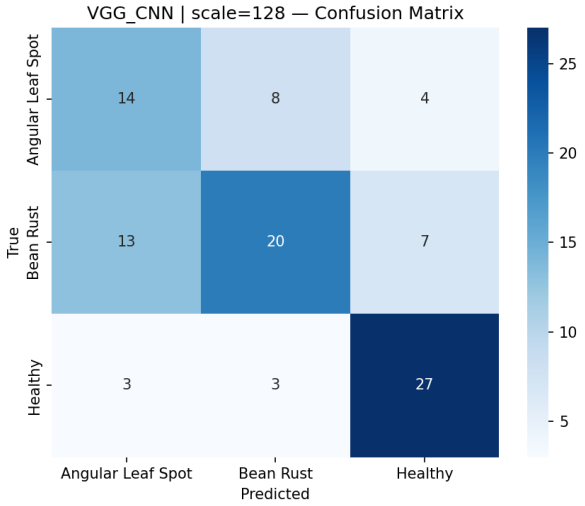
Figure 3 presents the confusion matrices of the VGG-style CNN evaluated at multiple input resolutions (32, 64, 128, and 224). As observed in Fig. 3, the model shows moderate performance at lower resolutions, with noticeable misclassification between Angular Leaf Spot and Bean Rust. Although there is slight improvement at higher resolutions, significant confusion persists, particularly for Angular Leaf Spot being predicted as Bean Rust. The Healthy class is consistently well-classified across all scales, indicated by strong diagonal values. However, the overall accuracy does not improve substantially with increasing resolution. Thus, Fig. 3 highlights the limited ability of the VGG model to leverage higher-resolution information effectively.



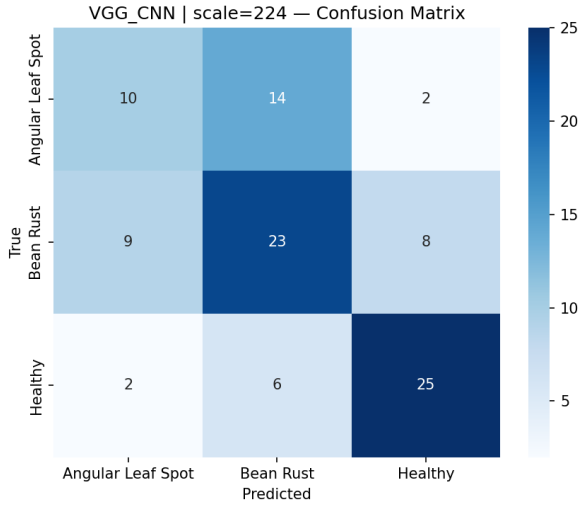
(a) VGG_32



(a) VGG_64



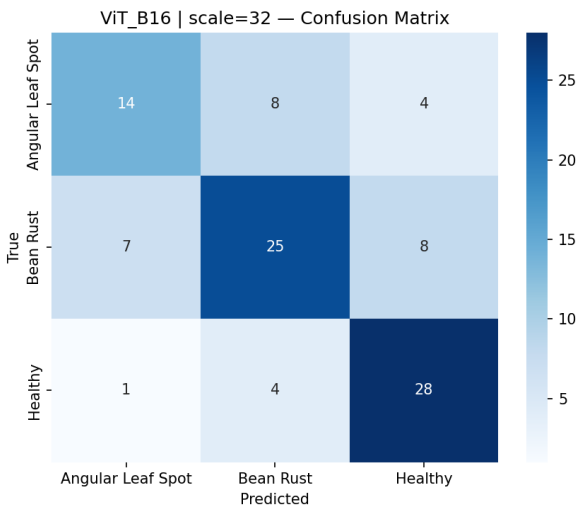
(c) VGG_128



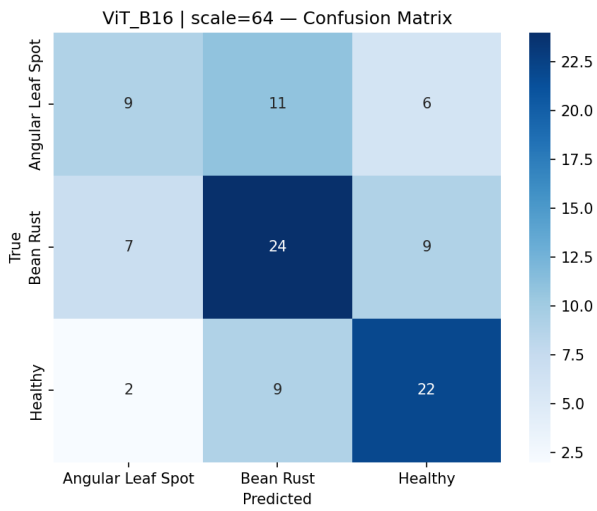
(d) VGG_224

Fig. 3 Confusion Matrix for VGG at different scales (a) VGG_32, (b) VGG_64, (c) VGG_128, and (d) VGG_224

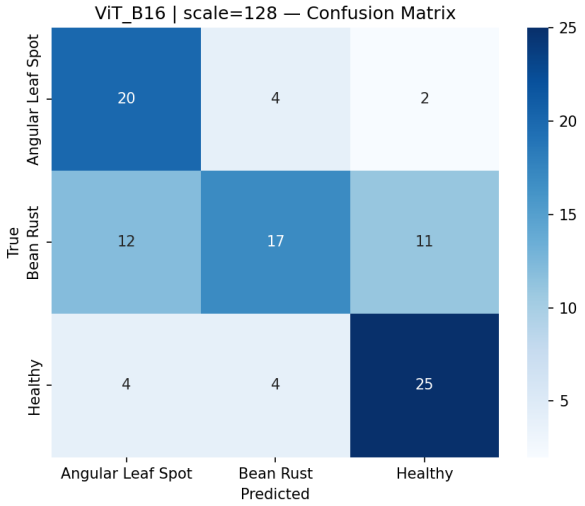
Figure 4 presents the confusion matrices of the ViT-B16 model evaluated across different input resolutions (32, 64, 128, and 224). As observed in Fig. 4, the model shows comparatively lower and less consistent performance across resolutions, especially at intermediate scales. At lower resolutions (32 and 64), there is noticeable confusion among all three classes, particularly between Angular Leaf Spot and Bean Rust. Although some improvement is observed at higher resolutions (128 and 224), misclassifications still persist, especially for the Bean Rust class, which is often confused with both Angular Leaf Spot and Healthy. The Healthy class maintains relatively stable performance across scales, but not as strongly as observed in CNN-based models. Overall, Fig. 4 indicates that the Vision Transformer is more sensitive to resolution variations and struggles to maintain consistent classification performance across multiple scales in this setup.



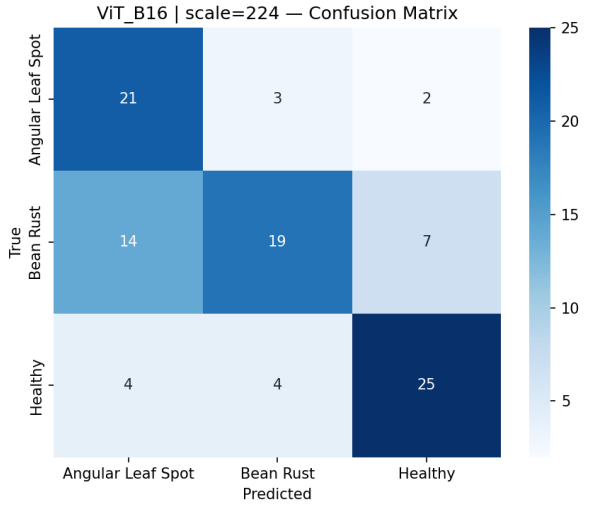
(a) ViT _32



(a) ViT _64



(c) ViT _128



(d) ViT _224

Fig. 4 Confusion Matrix for VGG at different scales (a) ViT_32, (b) ViT _64, (c) ViT _128, and (d) ViT _224

5.3. Accuracy Discussion

Figure 5 illustrates the test accuracy of the three models, VGG-style CNN, ResNet18, and ViT-B16, across different image resolutions (32, 64, 128, and 224). As observed in Fig. 5, ResNet18 consistently outperforms the other models at all scales, achieving the highest accuracy, with peak performance at higher resolutions (128 and 224). The VGG model demonstrates relatively stable but lower accuracy across all scales, showing limited improvement with increasing resolution. In contrast, ViT-B16 exhibits moderate performance, with noticeable variation across resolutions, particularly a drop at scale 64 followed by recovery at higher scales. Overall, Fig. 5 indicates that ResNet18 is the most robust model to resolution changes, while transformer-based approaches show sensitivity to scale variations in this setup.

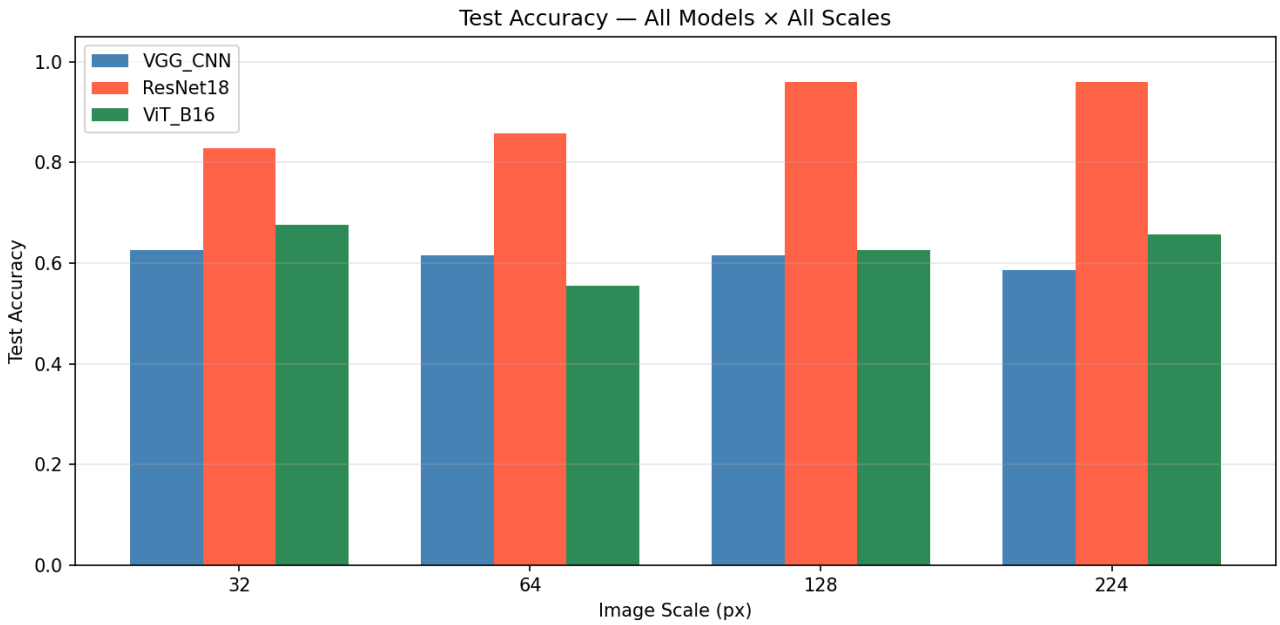


Fig. 5 Accuracy pertaining to three different models

DETAILED DISCUSSION

The results clearly show that the three models behave very differently across all four resolutions. ResNet18 stands out as the best performing model, VGG_CNN shows stable but limited performance, and ViT_B16 struggles throughout despite being the most advanced architecture. Each model tells a different story and each story has a clear and logical explanation.

Table 1. Test Accuracies based on all the scales for three different models

Model	Scale	Accuracy
VGG_CNN	scale= 32	0.6263
VGG_CNN	scale= 64	0.6162
VGG_CNN	scale=128	0.6162
VGG_CNN	scale=224	0.5859
ResNet18	scale= 32	0.8283
ResNet18	scale= 64	0.8586
ResNet18	scale=128	0.9596
ResNet18	scale=224	0.9596
ViT_B16	scale= 32	0.6768
ViT_B16	scale= 64	0.5556
ViT_B16	scale=128	0.6263
ViT_B16	scale=224	0.6566

Discussion 1: ResNet18: The Best Performing Model

ResNet18 is the strongest model in this experiment by a significant margin. At the highest resolution of 224×224, it achieves an accuracy of 95.96%, which means it correctly classifies

almost every test image. Even at the lowest resolution of 32×32, it still manages 82.83%, which is a very respectable result considering how little visual information a 32×32 image contains.

Why does ResNet18 perform so well?

The first and most important reason is that ResNet18 uses convolutional filters. These filters slide across the entire image looking for local patterns such as edges, color changes, spots, and textures. Bean disease classification is fundamentally a local feature detection problem. Angular Leaf Spot appears as small angular marks on the leaf. Bean Rust appears as scattered rust-colored pustules. A Healthy leaf has a uniform green surface. All of these distinguishing features are local in nature, and convolutional filters are perfectly designed to detect exactly these kinds of local patterns.

The second reason is the skip connections that ResNet is famous for. In a normal deep neural network, when information passes through many layers, it can get distorted or lost, this is called the vanishing gradient problem. ResNet solves this by adding shortcut connections that allow information to jump over layers and reach deeper parts of the network unchanged. This means ResNet can be deep enough to learn complex disease patterns without losing the simple low-level features like color and edges that are detected in the early layers.

The third reason is the benefit of ImageNet pretraining. ResNet18 was pretrained on over one million diverse images before we fine-tuned it on our bean dataset. During pretraining it already learned how to detect textures, edges, color gradients, and shapes. When we fine-tune it on bean disease images, it does not need to learn these basic visual concepts from scratch. It only needs to adapt its existing knowledge to focus on disease-specific patterns. With only around 1000 to 2000 bean images available, this pretrained knowledge gives ResNet18 a massive advantage.

What does the scale trend tell us?

ResNet18 improves as resolution increases, going from 82.83% at 32×32 to 95.96% at 128×128 and staying there at 224×224. This is the expected and healthy behavior. Higher resolution images contain more visual detail, and more detail means the convolutional filters can detect disease features more precisely. The fact that accuracy does not improve further from 128×128 to 224×224 suggests that 128×128 already contains enough detail for ResNet18 to make confident and correct decisions. This is an important finding, it means for this specific task, you do not always need full resolution images to achieve excellent performance.

Discussion 2: VGG_CNN as a Stable But Limited option

VGG_CNN is the custom model built from scratch in this project. It is a simple two-block convolutional network inspired by the VGG architecture. Its accuracy ranges from 58.59% to 62.63% across all four scales, which means it performs at a similar level regardless of the input resolution.

Why is VGG_CNN so consistent across scales?

The reason for this consistency is the AdaptiveAvgPool layer used in the model. This layer always compresses the feature maps to a fixed 4×4 output regardless of the input size. Whether the input image is 32×32 or 224×224, the model always produces the same sized feature representation before classification. While this makes the model resolution-independent, it also means that the extra detail available in higher resolution images is being compressed away and lost. The model cannot take advantage of the richer information in larger images.

Why is the accuracy low overall?

VGG_CNN was trained from scratch with no pretraining. It had to learn everything — basic edges, textures, colors, and disease-specific patterns — all from our small dataset of around 1000 to 2000 images. This is a very difficult task for any model. The model simply does not have enough training examples to properly learn the subtle differences between the three bean disease classes.

Additionally, the model has a relatively small capacity with only around 1.2 million parameters organized in just two convolutional blocks. While this makes it simple and easy to understand, it limits how complex the features it can learn are. The disease differences between Angular Leaf Spot, Bean Rust, and Healthy leaves require detecting fairly subtle visual patterns, and two convolutional blocks may not be deep enough to capture these patterns reliably.

What does the scale trend tell us?

Interestingly, VGG_CNN actually performs slightly worse at the highest resolution of 224×224 with 58.59% compared to 62.63% at 32×32. This seems counterintuitive but has a reasonable explanation. At higher resolutions, the images contain more detail and complexity. Without pretraining and with limited capacity, the model may actually get confused by the extra visual noise and fine details it does not know how to interpret. A simpler, lower resolution image with less detail is paradoxically easier for this limited model to classify.

The overall takeaway for VGG_CNN is that a custom model trained from scratch on a small dataset will always struggle, regardless of how the resolution changes. The bottleneck is not the resolution - it is the lack of pretrained knowledge and model capacity.

Discussion 3: ViT_B16: The Surprising Underperformer

ViT_B16 produces the most surprising and interesting results in this experiment. Despite being the most advanced and powerful architecture among the three, it performs poorly across all scales. Its accuracy ranges from just 55.56% at 64×64 to 67.68% at 32×32, and even at the full resolution of 224×224 it only reaches 65.66%. At 64×64, its accuracy of 55.56% is barely above random chance for a three-class problem, where random guessing would give around 33%.

This is a very important finding because it challenges the common assumption that a more advanced model always gives better results.

Why does ViT perform so poorly despite being a Transformer?

The core reason is a fundamental mismatch between what ViT was designed for and what our experiment asks it to do.

ViT works by cutting the input image into 196 fixed patches of 16×16 pixels each. Each patch is converted into a vector embedding and then all 196 embeddings are passed through a Transformer that uses Self-Attention to learn the relationships between patches. The final classification is made based on a special summary token called the CLS token.

This design is extremely powerful when the patches contain rich and meaningful visual information. However, bean disease features — spots, pustules, color patches — are small, local, and subtle. They may exist in only a few of the 196 patches. The Self-Attention mechanism, which is designed to find global relationships across all patches, is essentially looking for the wrong kind of information for this task. It is trying to understand how patch 1

relates to patch 150, when what actually matters is the local appearance of individual patches containing disease marks.

Why does ViT struggle especially at low resolutions?

At 32×32 and 64×64, the situation becomes even worse. Our code upsamples these small images to 224×224 before feeding them to ViT because ViT strictly requires 224×224 input. However, upsampling a 32×32 image to 224×224 does not recover any lost detail. The resulting image is simply a blurry, stretched version of the original with no new information.

When ViT then cuts this blurry image into 16×16 patches, every patch contains blurry, repetitive, and nearly identical pixel values. The patch embeddings are essentially meaningless vectors. The Self-Attention mechanism is then trying to find meaningful relationships between 196 patches that all look almost the same. This explains the dramatic drop to 55.56% at 64×64, the model is essentially operating blind.

Why does pretraining not save ViT here?

Even though we used a pretrained ViT-B/16 that was trained on ImageNet, the fine-tuning on our small dataset is not sufficient to adapt the model properly. ViT has approximately 86 million parameters. Fine-tuning 86 million parameters on only 1000 to 2000 bean images means there is simply not enough data to properly guide all those parameters toward the right solution. The pretrained knowledge from ImageNet, which involved distinguishing vastly different categories like dogs, cars, and aeroplanes, does not transfer as smoothly to the task of detecting subtle differences between three very similar looking leaf images.

ResNet18 with only 11 million parameters is much easier to fine-tune effectively on a small dataset because fewer parameters need adjustment and the convolutional inductive bias already points the model in the right direction.

What does the scale trend tell us about ViT?

The scale trend for ViT is irregular and inconsistent - 67.68% at 32×32, dropping to 55.56% at 64×64, recovering slightly to 62.63% at 128×128, and reaching its best at 65.66% at 224×224. This inconsistency itself is a finding. A robust model should show a smooth improvement as resolution increases, as ResNet18 does. ViT's irregular pattern shows that it is not reliably extracting useful features at any resolution, and its performance is somewhat random rather than systematically improving with more image detail.

Final Conclusion

This experiment delivers a clear and important message. ResNet18 dominates because its convolutional design naturally matches the local feature nature of bean disease detection, its skip connections preserve important information through deep layers, and its 11 million parameters are the right size to fine-tune effectively on a small agricultural dataset. VGG_CNN shows that training from scratch on a small dataset without pretraining will always produce limited results, regardless of resolution. The model is consistent but not accurate enough to be useful in practice. ViT_B16 demonstrates that architectural sophistication does not guarantee better results when the dataset conditions do not match the model's requirements. ViT needs millions of images, high resolution inputs with rich patch detail, and global feature relationships, none of which our bean disease dataset provides.

The broader lesson from these results is that **model selection should always be driven by the nature of the dataset and the task, not by the popularity or general power of the**

architecture. For small agricultural datasets with local disease features, a well pretrained convolutional model like ResNet18 remains the most practical and reliable choice.

6. CONCLUSION

This study explored cross-resolution consistency learning for robust bean disease classification using VGG-style CNN, ResNet18, and Vision Transformer (ViT-B16). The results demonstrate that incorporating consistency loss improves prediction stability across varying image resolutions. Among the models, ResNet18 achieved the highest accuracy and showed consistent improvement with increasing resolution, highlighting the effectiveness of convolutional architectures in capturing local disease features. In contrast, the VGG model exhibited stable but lower performance due to limited capacity and lack of pretraining, while ViT-B16 showed inconsistent results, reflecting its sensitivity to small datasets and reliance on global feature relationships. Overall, the findings emphasize that model performance depends not only on architectural complexity but also on alignment with the dataset and task characteristics. The study concludes that cross-resolution consistency learning enhances robustness, and pretrained CNN-based models remain the most reliable choice for small-scale agricultural image classification tasks.

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